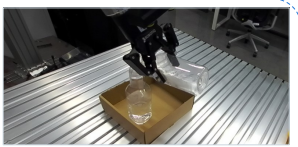
(a) Interaction across real & simulated worlds

Real



Pen operation



Pouring

Simulation



Push



Cover



Stack



Tool use

Images · measured state · execution feedback

Record

EEF poses + gripper

KnowBody

Images + state

(b) Experience-guided execution

Execution agent

One session per episode

Task



Insert key

Brief



Expose tip

Live state



EEF targets · action chunk



Free space: larger — Contact: finer

Query / return

Retrieve

Same-task process





…

Body / interaction knowledge



Body: marker
Rel.: EEF → tip
Cond.: pose

Check current conditions

On demand

Brief / evidence

Memory manager

Episode start

Prepare context

Review / learn

Source / scope checks

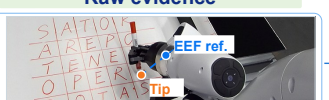
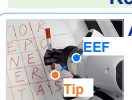
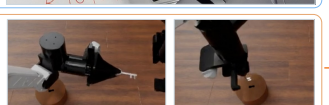
Append / revise

Read

Read

(c) Structured experience library

Organized by body, interaction relation and conditions

	Raw evidence	Conditional cases	Reusable knowledge
Body Geometry · tool mounting · visual action		 **Tool-dependent offset** EEF command — Tip (functional) — Tool-dependent offset — Schematic geometry	 **Act at the functional point** • Match the tool mounting • Recheck after regrasp
External Grasp · transform · interact		**Demonstration candidate**  Grasp layout — Insertion access	 **Preserve downstream access** • Keep the working end clear • Grasp · object · target geometry Candidate; check current grasp

Episode history  Stored with source links

Knowledge first · details on demand